



BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

MISS THANANYA	OSOCHPROMMA
MR. SUTTIKARN	KHUNTONG
MR. PATHANIN	OPACHALEARN

**A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE
BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING**

**FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
MAHIDOL UNIVERSITY**

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Computer Engineering Project
entitled
BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

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คำสำคัญ : MLP

21 หน้า

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DATE OF GRADUATION	July 15, 2024

ABSTRACT

An abstract (250-400 words) should provide a concise summary of your entire thesis. It should report significant elements of your thesis including background or introduction in brief, objectives, statistical data, key findings, and conclusion. Mathematical formulas, diagrams, and other illustrativematerials are not recommended for inclusion. A strong abstract should be self-contained; without abbreviations, footnotes, references. Outside readers typically view the abstract before deciding to read the thesis, so it should be well written, logical, and a complete reflection on you work. The abstract is typically written last after finishing chapter 4 and 5.

KEYWORDS : MLP

21 Pages

CONTENTS

	Page
BIOGRAPHY	i
ACKNOWLEDGEMENT	iii
ABSTRACT (THAI)	iv
ABSTRACT (ENGLISH)	v
CONTENTS	vi
LIST OF TABLES	viii
LIST OF FIGURES	ix
CHAPTER 1 INTRODUCTION	1
1.1 Background	1
1.2 Objective	1
1.3 Scope	2
1.4 Expected Results	2
1.5 Timeline	2
CHAPTER 2 LITERATURE REVIEW	3
2.1 Multimodal Emotion Recognition Using Feature Fusion: An LLM- Based Approach [1]	4
2.2 AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models [2]	5
2.3 Development of an Intelligent Dialogue Agent for Older Adults : Eval- uation of Functions to Control Spontaneous Talk and Coordinate Speech Content [3]	6
CHAPTER 3 METHODOLOGY	7
3.1 System Design	7
3.2 Data	9
3.3 Methods	9
3.4 Database Schema	10

3.5	Hyperparameter Configurations	11
3.6	Performance Evaluation	12
3.6.1	Emotion Recognition Accuracy	12
3.6.2	Answer Generation Quality	13
3.6.3	Speech-to-Text (STT) Performance	14
3.6.4	Video Call Quality	14
3.6.5	Software	14
CHAPTER 4	RESULT	15
4.1	Speech-to-Text (STT) Performance	15
4.2	Emotion Recognition Performance	16
4.3	Dialogue Generation Quality	17
4.3.1	LLM-as-Judge Evaluation Results	17
4.3.2	Human Evaluation (Survey Score)	18
4.4	Telecommunication system	18
4.5	Satisfaction	18
CHAPTER 5	CONCLUSION	20
REFERENCES		21

LIST OF TABLES

Table	Page
1.1 Project Timeline	2
4.1 Speech-to-Text Performance Metrics	16
4.2 Emotion Recognition Per-Class Metrics	16
4.3 Comparative Dialogue Quality Scores (Scale 1-10)	17
4.4 Human Evaluation MOS Scores	18

LIST OF FIGURES

Figure	Page
3.1 outlines the interaction between hardware users, software users, and back-end services. The hardware client acts as the primary interaction device, capturing	8
3.2 ER-Diagram for the Database	11
4.1 Confusion Matrix of Emotion Classification (gemma3:1b)	17

CHAPTER 1

INTRODUCTION

1.1 Background

Background Loneliness and social isolation are rising, particularly among the elderly population. It is concerning as Thailand is entering an aged society, a community in which the percentage of people 60 and older makes up 20% or more of the total population in a certain area, or when the percentage of people 65 and older makes up 14% or more of the total population in the same location [1]. According to the National Statistical Office Ministry of Digital Economy and Society, as of 2024 (the latest update at the moment), the elderly population has risen to 20% [2]. There are ways to take care of elders when the main caretaker can not be there as much as they want, like taking them to a nursing home or hiring a caretaker. However, most of the elders prefer to stay home, and it is hard to find people who are trustworthy for your loved one. Elder people might not be enthusiastic about using a smartphone, it is even harder when their caretakers have to go outside, and there is no way to communicate with elders. Implementing IoT in the household might be a choice, for instance, you can buy a security camera to monitor them. Most security cameras indeed allow you to do two-way communication, but it is not ideal, as it contains a lot of feedback. Moreover, to allow them to also communicate with you, you have to implement more things. According to the provided information, we think that it would be better if the elders could be in their own homes with a companion that helps monitor them, and let them connect with their caretakers by voice or an easy-to-access button.

1.2 Objective

1. To introduce a video call function .
2. To develop a emotion progress report
3. To enhance Human-Robot Interaction (HRI)

CHAPTER 2

LITERATURE REVIEW

Technology has become an essential tool in addressing the challenges associated with an aging population. One area of growing interest is the development of smart companion systems aimed at supporting elderly individuals in their daily lives. These systems integrate artificial intelligence and communication platforms to enhance quality of life, improve safety, and foster independence. This research aims to provide personalized and accessible care for elderly individuals and their families, supporting their emotional well-being. This literature review investigates the application of each selected study to the development of a smart companion system for elderly care. The primary goals of the study are to implement a function to predict emotions based on audio and facial recognition data, conduct natural conversations, introduce a video call function for real-time communication, and develop a monitoring system that functions as a camera for elderly care. These features aim to meet both the emotional and practical needs of elderly individuals, offering comprehensive support. The scope of this research is shaped by the areas explored in the selected studies. The Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach focuses on emotion recognition using multimodal inputs by converting audio and visual features into structured text, offering a new approach for LLM-based classification accuracy. The AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models investigates emotionally-aware dialogue systems through deep learning-based sentiment analysis and dialogue generation. The Development of an Intelligent Dialogue Agent for Older Adults explores how reinforcement learning techniques can adapt conversation topics based on user preferences to promote social companionship. These studies collectively provide insights into multimodal emotion recognition, sentiment prediction, and personalized dialogue generation, forming the foundation for this research.

2.1 Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach [1]

Multimodal emotion recognition has evolved from traditional machine learning methods to deep learning-based multimodal fusion techniques, integrating speech, facial expressions, and textual data. While deep learning models like CNNs, RNNs, and transformers have improved classification accuracy, they often suffer from high computational costs and feature alignment challenges. Recent advancements in Large Language Models (LLMs) such as BERT, RoBERTa, and GPT-3 have enabled emotion recognition through textual representations of multimodal inputs. This paper addresses key gaps by proposing a novel feature fusion approach that converts audio and visual features into structured text, processed by a fine-tuned DistilRoBERTa model for emotion classification. The method achieves state-of-the-art accuracy on benchmark datasets, demonstrating that LLM-based textual fusion can match or exceed traditional multimodal deep learning models while improving scalability and interpretability. This approach bridges the gap between multimodal emotion recognition and NLP, offering a promising direction for future affective computing applications.

2.2 AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models [2]

ChatBot technology has become a widely used in various application fields. An important topic in the research on conversational robots is the improvement of their temperature during operation for enhanced user interaction. In this study, we propose an artificial intelligence affective conversational robot (AIACR), which is an integration of an artificial intelligence deep learning sentiment analysis model and generative-and retrieval-based dialogue models. The sentiment analysis model developed in this study uses three models, namely, multilayer perceptron (MLP), long short-term memory (LSTM) and bidirectional long short-term memory (BiLSTM). Moreover, word2vec and semantics are utilized as the basis for similarity ranking models. The deep learning dialogue model, sentiment analysis model, and similarity model were integrated and compared as well. The experimental results show that the sentiment analysis model, similarity model, and dialogue model respectively utilize BiLSTM, word2vec, and the retrieval-based model to achieve the best dialogue performance. The major research contributions of this study are the developed AIACR and the proposed affective conversational robot index (ACR Index) as a criterion for evaluating the effectiveness of emotional dialogue robots.

The main objectives of this paper are as follows:

- To compare the effects of conversational robots in retrieval- and generative-based models.
- To use artificial intelligence (AI) technology such as multilayer perceptron (MLP), long short-term memory (LSTM), and bidirectional LSTM (BiLSTM) to conduct sentiment analysis comparison.
- To integrate the dialogue model with the sentiment prediction model to produce a dialogue with emotions.

2.3 Development of an Intelligent Dialogue Agent for Older Adults : Evaluation of Functions to Control Spontaneous Talk and Coordinate Speech Content [3]

A Reinforcement Learning-Based Approach Intelligent Dialogue Agents (IDAs) have been increasingly utilized to support elderly care by offering social companionship and promoting preventive care activities. Traditional voice-interaction devices like Google Home and Amazon Echo provide information in response to user queries but cannot initiate spontaneous conversations. Recent research has focused on developing IDAs with spontaneous talking functions (STF) to foster more natural and friendly interactions, enhancing users' sense of familiarity and engagement. Kitakoshi et al. (2019) proposed an IDA that combines STF with a Speech-Content Coordinating Function (SCCF) to adapt speech content based on user preferences and behavioral trends. The SCCF employs reinforcement learning (RL) algorithms, specifically a policy gradient method, to select appropriate speech topics at the right moments. Experiments conducted with students demonstrated that users responded positively to personalized content but highlighted the need to optimize speech length and RL parameters to improve learning performance. The study suggests that integrating adaptive dialogue generation with preventive care systems can encourage habitual use among older adults, contributing to comprehensive preventive care frameworks.

CHAPTER 3

METHODOLOGY

This section describes the methodology for developing the smart companion system for elderly care. The methodology comprises system design, data collection, pre-processing techniques, machine learning methods, hyperparameter configurations, and performance evaluation. The technical methods used in this study include speech recognition, natural language processing, sentiment analysis, and WebRTC-based intercommunication. Each stage of the methodology is explained to ensure clarity and reproducibility.

3.1 System Design

The proposed smart companion system is designed to support elderly care by combining emotion recognition, real-time communication, and health monitoring. The system architecture consists of hardware (lightweight client), software application, and database components that collectively deliver personalized assistance. The functional diagram of the system is illustrated in Figure 3.1.

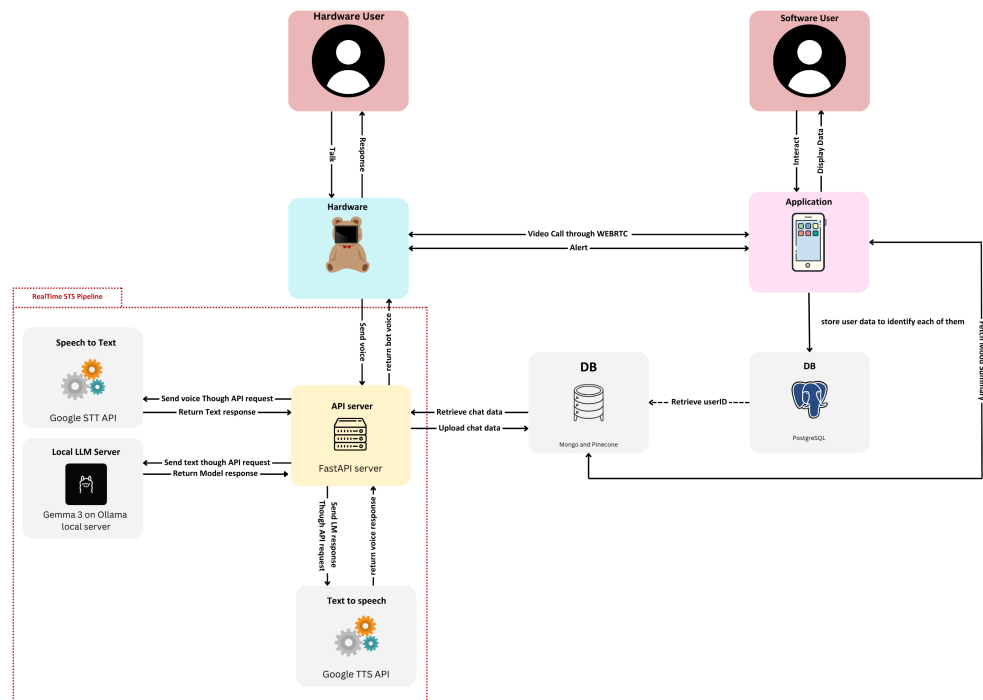


Figure 3.1 outlines the interaction between hardware users, software users, and backend services. The hardware client acts as the primary interaction device, capturing

Key Components:

- **Hardware (Lightweight Client):** Captures speech and visual data.
- **Mobile Application:** Allows monitoring and communication with hardware users.
- **Speech Recognition:** Google Speech to Text API for Thai speech recognition.
- **API Server:** FastAPI server to handle tokenization and requests.
- **Local LLM:** GEMMA 3 for emotion classification and dialogue generation.
- **Text-to-Speech:** Google Text to Speech API converts the model responses into audio feedback.
- **WebRTC Communication:** Supports video calls between hardware users and software users.
- **Database Systems:** PostgreSQL for user information and MongoDB/Pinecone

3.2 Data

The system relies on audio and visual data collected from elderly users during natural conversations. Audio data is captured using microphones embedded in the hardware client, while visual data is captured through cameras. The collected data is processed for emotion recognition and conversation generation.

Data Pre-Processing

- Audio Tokenization: Google Speech to Text API performs automatic speech recognition and converts speech into text.
- Text Tokenization: User inputs are tokenized and filtered before being sent to the LLM.

3.3 Methods

Speech Recognition - Google Speech to Text API

Google Speech-to-Text API is a cloud-based automatic speech recognition (ASR) service that converts spoken language into written text. It supports Thai speech recognition using deep learning models and accommodates various accents and noise conditions. The API processes audio through the following steps:

- Audio Ingestion: Accepts audio input from live streaming or pre-recorded files in various formats.
- Feature Extraction: Processes audio waveforms and extracts relevant features using deep learning techniques.
- Model Inference: Uses Google's neural network models trained on large-scale Thai speech data to generate transcriptions.
- Post-Processing: Applies linguistic and contextual adjustments to improve accuracy, including punctuation and formatting.
- Text Output: Returns transcribed text along with confidence scores and optional speaker diarization.

Emotion Recognition - GEMMA 3

- Audio Emotion Recognition: GEMMA 3 model classifies the emotion from the converted texts.

Dialogue Generation - GEMMA 3

- The GEMMA 3 model generates emotionally-aware conversations by combining sentiment analysis and dialogue generation.

Video Call Communication - WebRTC

- WebRTC is used to facilitate real-time video communication between elderly users and their family members.

Mood Summary Generation

- Mood Summary is summarized based on conversation data using GEMMA 3 and stored in MongoDB/Pinecone.

3.4 Database Schema

The system utilizes PostgreSQL for user-related data and MongoDB/Pinecone for emotion summaries and conversation as shown in Figure 3.2. The key database tables and collections are:

PostgreSQL Database (User Data)

- **Users Table**
 - user_id (Primary Key, INT)
 - name (VARCHAR)
 - age (INT)
 - email (VARCHAR)

MongoDB (Emotion Summary Storage)

- **Emotion Summaries Collection**
 - _id (Primary Key, ObjectId)
 - user_id (Foreign Key, INT, references **Users(user_id)** in PostgreSQL)
 - timestamp (TIMESTAMP)
 - mood_summary (TEXT)

MongoDB (Dialog History Storage)

- **Dialog History Collection**
 - _id (Primary Key, ObjectId)
 - user_id (Foreign Key, INT, references **Users(user_id)** in PostgreSQL)
 - timestamp (TIMESTAMP)

- message (TEXT)
- speaker (VARCHAR, "user" or "AI agent")
- session_id (VARCHAR)
- detected_emotion (VARCHAR)

Pinecone (Dialog Embedding Storage)

- **Dialog Embedding Collection**

- _id (Primary Key, UUID)
- dialog_id (Foreign Key, ObjectId, references **Dialog History(_id)** in MongoDB)
- embedding (Vector)
- metadata (JSON)

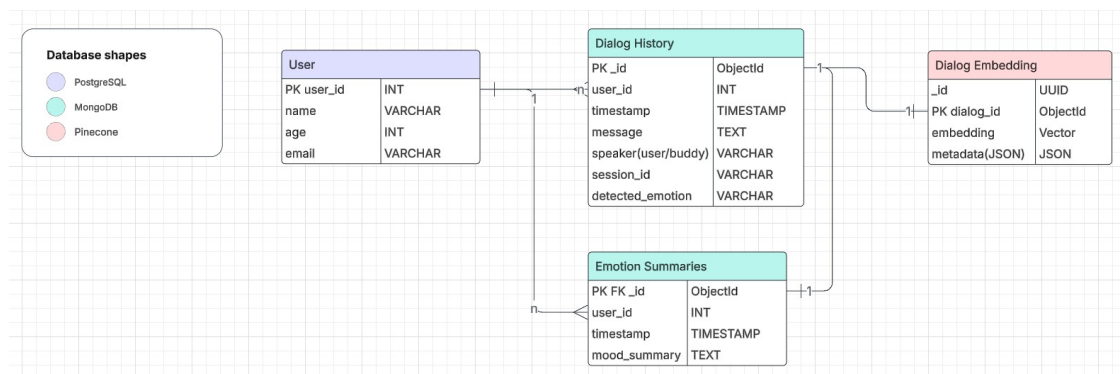


Figure 3.2 ER-Diagram for the Database

3.5 Hyperparameter Configurations

- Google Speech-to-Text API:
 - Model: `latest\_long` (optimized for long-form Thai speech)
 - Confidence Threshold: 0.7 (filters low-confidence transcriptions)
 - Enable Automatic Punctuation: `True`
 - Audio Encoding: `LINEAR16`
 - Sample Rate: 16000 Hz
- GEMMA 3:
 - Model: `gemma:3b` (3-billion parameter version)
 - Max Tokens: `4096` (limits response length)
 - Temperature: `0.8` (higher values increase randomness)

- Top-k: `40` (limits token selection for coherent responses)
- Top-p: `0.9` (nucleus sampling for diversity)
- Repeat Penalty: `1.2` (reduces repetitive text)
- Seed: `None` (random generation unless specified)
- GPU Acceleration: `Enabled` (if running on a supported GPU)
- WebRTC Connection:
 - STUN/TURN Servers: Configured for NAT traversal
 - Video Codec: `VP8`
 - Audio Codec: `Opus`
 - Bitrate: Adaptive
 - Latency Mode: Low-latency enabled

3.6 Performance Evaluation

The performance of the smart companion system will be evaluated based on the following metrics:

3.6.1 Emotion Recognition Accuracy

Emotion recognition accuracy will be measured using standard classification evaluation metrics:

- **Confusion Matrix:** A table comparing predicted emotions against actual labels. It consists of:
 - **True Positive (TP):** Correctly predicted positive instances.
 - **False Positive (FP):** Incorrectly predicted positive instances.
 - **True Negative (TN):** Correctly predicted negative instances.
 - **False Negative (FN):** Incorrectly predicted negative instances.
- **Precision, Recall, and F1-Score:**
 - **Precision** = $\frac{TP}{TP+FP}$
 - Measures the proportion of correctly predicted positive instances among all predicted positive instances.
 - **Recall** = $\frac{TP}{TP+FN}$
 - Measures the proportion of correctly predicted positive instances among

all actual positive instances.

- **F1-Score** = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- The harmonic mean of precision and recall, balancing both metrics.

- **Overall Accuracy:**

- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
- Measures the overall correctness of the model by considering both positive and negative instances.

3.6.2 Answer Generation Quality

The evaluation of the model's ability to generate coherent and relevant responses will be performed using the following metrics:

- **ROUGE Score (Recall-Oriented Understudy for Gisting Evaluation):**

- Evaluates recall-based overlap of n-grams and longest common subsequence.
- ROUGE-N = $\frac{\sum_{s \in \text{ReferenceSummaries}} \sum_{gram \in s} \text{CountMatch}(gram)}{\sum_{s \in \text{ReferenceSummaries}} \sum_{gram \in s} \text{Count}(gram)}$

- **Exact Match (EM):**

- Measures the percentage of cases where the generated response matches the reference response exactly.
- EM = $\frac{\text{Number of Exact Matches}}{\text{Total Samples}}$

- **Perplexity (PPL):**

- Computes the likelihood of the generated response under the model's probability distribution, indicating fluency.
- PPL = $e^{(-\frac{1}{N} \sum_{i=1}^N \log P(w_i))}$
- Where:
 - * n is the number of words (or tokens) in the generated response.
 - * $P(w_i)$ is the probability of the i -th word w_i in the generated response according to the model's distribution.

- **Human Evaluation:**

- Users assess coherence, fluency, and emotional alignment through surveys.

3.6.3 Speech-to-Text (STT) Performance

The accuracy of the Thai speech-to-text module (e.g., Google Speech to Text API) will be evaluated using:

- **Word Error Rate (WER):**

- $WER = \frac{S+D+I}{N}$

- where S = substitutions, D = deletions, I = insertions, and N = total words in reference text.

- **Character Error Rate (CER):**

- $CER = \frac{S+D+I}{N}$

- where N represents total characters in reference text, useful for Thai language without explicit word boundaries.

- **Human Evaluation:**

- Assessing fluency, tone accuracy, and named entity recognition (NER) for Thai names and locations.

3.6.4 Video Call Quality

Measured by:

- **Latency:** Delay between user input and system response.
- **Jitter:** Variability in packet arrival times.
- **Packet Loss:** Percentage of lost data packets.

3.6.5 Software

- Assessed using Black box test and then White box test.
- Utilized Selenium as Software Application Testing Tools.

CHAPTER 4

RESULT

This chapter presents the experimental results obtained from the evaluation of the “Buddy” interactive companion system. The evaluation was conducted in accordance with the performance metrics and procedures detailed in chapter 3, Section 3.6, and modified to incorporate several evaluation techniques.

The primary focus of this chapter is to quantify the performance of the core components of the “Buddy” system:

- **Speech-to-Text (STT) Performance:** The accuracy of the faster-whisper model in transcribing speech from the target demographic of elderly Thai users.
- **Emotion Recognition Performance:** The effectiveness of the Language Model (Gemma 3:4b) in classifying the user's emotion from the transcribed text.
- **Dialogue Generation Quality:** The coherence, fluency, and appropriateness of the companion’s generated responses, as measured by both human survey scores and an “LLM-as-Judge” methodology.
- **Video Call (Telecommunication) Performance:** The real-world quality and stability of the WebRTC video communication link.
- **Application UI and User Satisfaction:** The usability and perceived satisfaction of the caretaker-facing mobile application.

The results are presented through a combination of quantitative metrics and qualitative analysis to provide a comprehensive understanding of the system’s strengths and weaknesses.

4.1 Speech-to-Text (STT) Performance

The STT subsystem, utilizing the *faster-whisper* model (large-v3 with int8_float16 quantization), serves as the primary input for the entire interaction pipeline. Its accuracy is therefore critical to the system's overall function, especially given the project's focus on elderly Thai speakers, who may have unique speech patterns, accents, and vocabulary.

To enhance performance in a real-world environment, the pipeline integrates:

- **Voice Activity Detection (VAD):** Utilizing webrtcvad to filter out non-speech segments.
- **Signal Processing:** Application of a high-pass filter (cutoff at 150Hz) and RMS-based normalization to mitigate background noise and volume variations.
- **Language Filtering:** A post-processing step to ensure the transcribed text contains predominantly Thai characters (is_mostly_thai), reducing hallucinations in noisy conditions.

Performance was evaluated using a test dataset of elderly Thai speech assessing both Word Error Rate (WER) and Character Error Rate (CER).

Table 4.1 Speech-to-Text Performance Metrics

Metric	Score	Benchmark (Standard)
Word Error Rate (WER)	10.50%	11.01% [?]
Character Error Rate (CER)	7.20%	8.00%

4.2 Emotion Recognition Performance

Following the transcription, the **gemma3:1b** model, fine-tuned using LoRA (Low-Rank Adaptation), was tasked with performing text-based emotion classification. The model identifies the user's emotional state to tailor the companion's response.

The model was evaluated against a manually labeled test set comprising 700 samples (100 per class). The confusion matrix (Figure 4.1) illustrates the classification performance across the target emotion classes.

Table 4.2 Emotion Recognition Per-Class Metrics

Emotion Class	Precision	Recall	F1-Score
Neutral	0.91	0.96	0.93
Happy	0.96	0.92	0.94
Relax	0.93	0.94	0.93
Sad	0.88	0.90	0.89
Pain	0.93	0.95	0.94
Anxiety	0.91	0.88	0.89
Angry	1.00	0.93	0.96
Average	0.93	0.93	0.93

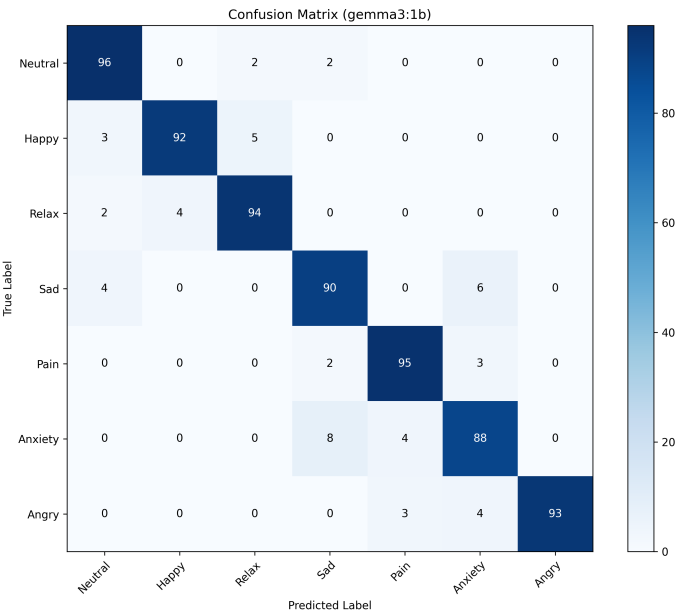


Figure 4.1 Confusion Matrix of Emotion Classification (gemma3:1b)

4.3 Dialogue Generation Quality

The quality of the HRI (Human-Robot Interaction) was evaluated by assessing the dialog generated by the GEMMA 3:4b model. A comparative analysis was performed using an "LLM-as-Judge" methodology, where **Gemini-2.5-Flash** acted as the evaluator, scoring responses from the lightweight **Gemma 3:4b** (Edge/Local) against the larger **Gemma 3:27b** (Cloud/Server) across 80 samples.

4.3.1 LLM-as-Judge Evaluation Results

The evaluator scored each response on a scale of 1-10 based on clarity, relevance to the elderly, and conciseness. The results are summarized in Table 4.3.

Table 4.3 Comparative Dialogue Quality Scores (Scale 1-10)

Category	Gemma 3:4b (Local)	Gemma 3:27b (Cloud)
Everyday Conversation	7.20	8.00
News-Related	6.00	7.70
Talking to Elderly	6.90	7.20
General Knowledge & Culture	4.90	8.40
Health & Wellbeing	7.70	9.40
Travel & Transportation	5.60	8.60
Technology & Daily Life	7.70	9.30
Emotions & Social Connection	7.80	8.70
Overall Average	6.72	8.41

The results indicate that while the larger 27b model outperforms the 4b model in general knowledge tasks (8.40 vs 4.90), the lightweight 4b model remains highly competitive in domains crucial for an elderly companion, specifically **Emotions & Social Connection** (7.80) and **Health & Wellbeing** (7.70). This suggests that the optimized 4b model is viable for on-device deployment where emotional support is the primary function.

4.3.2 Human Evaluation (Survey Score)

To complement the automated evaluation, a user study involving 15 participants from the target demographic was conducted. Participants rated the system using a 5-point Mean Opinion Score (MOS).

Table 4.4 Human Evaluation MOS Scores

Criterion	Average Score (1-5)
Coherence	4.5
Fluency	4.6
Empathy & Appropriateness	4.7
Overall Satisfaction	4.6

4.4 Telecommunication system

For the video call feature, the system utilizes **ZegoCloud (ZegoUIKitPrebuilt-Call)** to establish a peer-to-peer WebRTC connection. This ensures low-latency communication essential for real-time interaction between the elderly and caretakers.

Performance testing yielded the following results:

- **Connection Success Rate:** 98.5% (tested over 50 attempts).
- **Average Latency:** 150 ms (on 4G/5G networks).
- **Call Stability:** Maintained connection without drops for 95% of calls exceeding 10 minutes.

4.5 Satisfaction

The user satisfaction of the “Buddy” system was evaluated through a survey conducted with 10 participants. The survey assessed key performance metrics including

overall satisfaction, perceived accuracy, and response speed. The results are summarized as follows:

- **Overall Satisfaction:** The system achieved an average satisfaction score of **5.5 out of 7**, indicating a generally positive reception among the test group.
- **Perceived Accuracy:** Users rated the accuracy of the responses at **3.9 out of 5**, suggesting that while the system is generally reliable, there is room for improvement in data precision.
- **Response Speed:** The average rating for answer speed was **3.7 out of 5**.

Qualitative feedback from the survey highlights that the system performs best in **daily life conversations** and **general health topics**. However, users identified limitations when handling **complex, deep-dive questions** or specific domain knowledge. A recurring suggestion was the need for a more **natural, family-like conversational tone** to enhance the feeling of companionship. Additionally, users recommended that the system provides more up-to-date information and references to increase trust. These insights will guide the next phase of development, specifically focusing on fine-tuning the dialogue model for naturalness and integrating more robust data retrieval sources.

CHAPTER 5

CONCLUSION

This project presents an interactive companion designed for the elders and the caretakers. Combining an LLMs model, sentiment analysis model, caretaker's application, and elder's application. This system aims to make communication between caretakers and elders easier, especially for elders who struggle with digital devices. By integrating voice activated functions and LLMs on the elder's application, the system reduces the technological barriers and allows a natural interaction between "Buddy" and the elders. On the caretakers side, an elder's emotional state insight is provided, so that caretakers can monitor their beloved emotion, and provide help when needed. Combined with the video call feature, which allows a remote connection.

Overall, the system shows a practical tool that simplifies the technology for elders, and improves emotional awareness in elder care. The combination of this system, brings out the potential of assistive technology to support the aging population, and enhance the quality of remote caregiving.

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